

Table 1. **Coverage of the 39 closest prior surveys:** every verified survey found by the search or by citation searching whose topic is calibration and uncertainty, reinforcement learning for language models, closed-set decisions, acting on confidence, hallucination or conformal prediction. CAL confidence estimation and calibration; RLC calibration-aware training of any kind; RLR reinforcement learning whose reward is a calibration or proper-scoring-rule objective; DEC decisions over a predefined option set; TYP typed, per-option decision outputs; ESC acting on confidence (abstention, deferral, routing); HAL hallucination. ✓ covered, ~ partial, – absent, ? not determinable. The RLR marks were checked in the full text or full reference list of every survey; the other marks come from abstracts and contents, corrected from a full-text reading for the surveys marked †. A **A** marks a column that no prior survey fully covers (TYP). Each full RLR mark is one subsection scoped to a single slice: 2024 methods [Li et al. 2025e], LLM agents [Lin et al. 2026] and forecasting [Xu et al. 2026b]. The last row marks the dimensions within this survey’s scope (●); hallucination is in scope only through abstention training and the theory works (○).

Survey	Venue	CAL	RLC	RLR	DEC	TYP	ESC	HAL
<i>Calibration and uncertainty quantification</i>								
Geng et al. [2024a]†	NAACL	✓	~	–	✓	–	~	~
Liu et al. [2025a]†	KDD	✓	~	–	–	–	~	~
Shorinwa et al. [2025]†	CSUR	✓	~	~	–	–	~	✓
Xia et al. [2025]	ACL Findings	✓	–	–	–	–	–	~
Kang et al. [2025a]†	arXiv	✓	–	–	–	–	~	✓
Zhang et al. [2026a]†	arXiv	✓	~	–	–	–	~	~
Wang [2023]†	arXiv	✓	–	–	✓	–	~	–
Li et al. [2025e]†	TMLR	✓	✓	✓	?	–	~	~
Lin et al. [2026]†	arXiv	✓	~	✓	?	~	✓	?
Li et al. [2025g]	ACL	✓	~	~	?	–	~	~
Xie et al. [2024b]	arXiv	✓	–	~	?	–	?	–
Abbasli et al. [2025]	IEEE TNNLS	✓	?	~	?	–	?	~
Huang et al. [2024c]	arXiv	✓	?	–	?	–	?	?
Beigi et al. [2024b]	arXiv	✓	?	–	?	–	?	?
Oh et al. [2026]	ACL	✓	~	–	?	–	~	?
Xia et al. [2026]	arXiv	✓	–	–	?	–	?	?
Liu et al. [2026b]	arXiv	✓	~	~	?	–	?	?
Sun et al. [2026b]	arXiv	✓	–	–	?	–	?	?
Xu et al. [2026b]†	arXiv	~	✓	✓	~	–	–	–
Ulmer et al. [2025]†	arXiv	✓	~	~	–	–	–	–
Zhang and Wang [2026]	J. Comput. Sci. Technol.	✓	?	–	~	–	?	?
He et al. [2026b]	Inf. Fusion	✓	?	–	–	–	?	?
<i>Reinforcement learning for LLMs</i>								
Zhang et al. [2025f]†	arXiv	–	–	~	–	–	–	–
Liu et al. [2025c]†	arXiv	–	–	–	–	–	–	–
Kaufmann et al. [2023]†	TMLR	–	?	–	–	–	–	–
Yu et al. [2026a]†	arXiv	–	?	–	–	–	–	–
<i>Closed-set decisions and judging</i>								
Gu et al. [2024]	arXiv	~	–	–	✓	–	–	–
Li et al. [2024c]	arXiv	~	–	~	✓	–	–	–
Li et al. [2024b]	arXiv	~	–	–	✓	–	–	–
<i>Acting on confidence (abstain, defer, route)</i>								
Wen et al. [2025]†	TACL	~	~	~	–	–	✓	~
Hendrickx et al. [2024]	Mach. Learn.	~	–	–	✓	–	✓	–
Strong et al. [2025]	Zenodo (preprint)	~	–	–	✓	–	✓	–
Moslem and Kelleher [2026]†	TMLR	~	–	–	–	–	✓	–

Table 1, continued

Survey	Venue	CAL	RLC	RLR	DEC	TYP	ESC	HAL
Boudiaf et al. [2026] [†]	arXiv	✓	—	~	~	—	✓	?
<i>Hallucination</i>								
Huang et al. [2025d]	ACM TOIS	~	—	—	—	—	—	✓
Alansari and Luqman [2026]	Comput. Sci. Rev.	~	—	~	—	—	—	✓
Zhang et al. [2025a] [†]	arXiv	~	~	~	—	—	—	✓
<i>Conformal prediction</i>								
Campos et al. [2024] [†]	TACL	✓	—	—	✓	—	~	—
Ashby et al. [2026]	Phil. Trans. A	✓	—	—	~	—	~	—
Surveys covering (✓)		24	2	3	8	0	6	5
This survey (scope)		●	●	●	●	●	●	○